Problem 1: (1) $X \sim N(n, p)$

We need to find p s.t. $\mathbb{P}(X=k)$ for $k=0,1\cdots,n$.

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \binom{n}{k} \left(\frac{p}{k}\right)^k \left(\frac{1-p}{n-k}\right)^{n-k} \cdot k^k (n-k)^{n-k}$$

$$= \binom{n}{k} k^k (n-k)^{n-k} \left[\frac{p}{k}\right]^k \left(\frac{1-p}{n-k}\right)^{n-k}$$

$\because \left(\frac{p}{k}\right) \cdot k + \left(\frac{1-p}{n-k}\right) \cdot (n-k) = 1$

$\therefore \left(\frac{p}{k}\right)^k \left(\frac{1-p}{n-k}\right)^{n-k} \leq \left(\frac{1}{n}\right)^n$

This equals when $\frac{p}{k} = \frac{1-p}{n-k}$, which means $p = \frac{k}{n}$.

(2) We can use the result from (1), which gives $p = \frac{8}{8+4} = \frac{2}{3}$

Problem 2:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\,\mathbb{P}(X=k)$$

$$= \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!}$$

$$= \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-1)!}$$

$$= e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

$\because \sum_{k=0}^{\infty} \mathbb{P}(X=k) = 1$

$\therefore \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1$

$\therefore e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = 1$

$\therefore \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$

$\therefore \mathbb{E}[X] = e^{-\lambda} \cdot \lambda \cdot e^{\lambda} = \lambda$